



# MALLA REDDY UNIVERSITY

**Department of Computer Science and Engineering**

**Subject Name: Software Quality Testing**

**Subject Code: MR22-1CS0164**

**Year & Semester: IV-I**

**Unit-Wise Question Bank**

| <b>Q.no.</b> | <b>Questions (Unit-1)</b>                                                                                                  | <b>Marks</b> | <b>Section</b> | <b>Unit</b> |
|--------------|----------------------------------------------------------------------------------------------------------------------------|--------------|----------------|-------------|
| 1            | Define Software Quality. Why is it important in SDLC?                                                                      | 8            | Section-1      | Unit-1      |
| 2            | Explain Cost of Quality (COQ) and its components?                                                                          | 8            | Section-1      | Unit-1      |
| 3            | Compare: Quality vs Reliability vs Security vs Maintainability?                                                            | 8            | Section-1      | Unit-1      |
| 4            | Describe McCall's Quality Model with its categories?                                                                       | 8            | Section-1      | Unit-1      |
| 5            | Compare McCall's Model with Boehm's Model (1978) and provide examples?                                                     | 8            | Section-1      | Unit-1      |
| 6            | Explain CMMI and its levels?                                                                                               | 8            | Section-1      | Unit-1      |
| 7            | Differentiate between SQA, Quality Control (QC), and Testing?                                                              | 8            | Section-1      | Unit-1      |
| 8            | What are the roles and skills required for an SQA team?                                                                    | 8            | Section-1      | Unit-1      |
| 9            | How does SQA impact software maintenance and reliability?                                                                  | 8            | Section-1      | Unit-1      |
| 10           | Why is prevention better than detection in software quality?                                                               | 8            | Section-1      | Unit-1      |
| <b>Q.no.</b> | <b>Questions (Unit-2)</b>                                                                                                  | <b>Marks</b> | <b>Section</b> | <b>Unit</b> |
| 1            | Explain the Test Plan Format with its components?                                                                          | 8            | Section-2      | Unit-2      |
| 2            | Define a Test Scenario and provide an example?                                                                             | 8            | Section-2      | Unit-2      |
| 3            | What is a Test Case? Write an example for ATM withdrawal?                                                                  | 8            | Section-2      | Unit-2      |
| 4            | Differentiate between Positive and Negative Test Cases?                                                                    | 8            | Section-2      | Unit-2      |
| 5            | Write 3 positive and 3 negative test cases for user registration?                                                          | 8            | Section-2      | Unit-2      |
| 6            | List and explain 5 key points to be considered while creating a test plan?                                                 | 8            | Section-2      | Unit-2      |
| 7            | What is the importance of Suspension and Resumption Criteria in testing?                                                   | 8            | Section-2      | Unit-2      |
| 8            | Explain how Test Deliverables are documented in a test plan. Give examples?                                                | 8            | Section-2      | Unit-2      |
| 9            | Explain the significance of the "Features Not to be Tested" section in a Test Plan?                                        | 8            | Section-2      | Unit-2      |
| 10           | Create a Sample test plan for an E-Commerce Website.                                                                       | 8            | Section-2      | Unit-2      |
| <b>Q.no.</b> | <b>Questions (Unit-3)</b>                                                                                                  | <b>Marks</b> | <b>Section</b> | <b>Unit</b> |
| 1            | What is Boundary Value Analysis (BVA) in software testing, and what is its primary objective? And why is BVA considered an | 8            | Section-3      | Unit-3      |

|              |                                                                                                                                                                                           |              |                |             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------|-------------|
|              | effective test design technique?                                                                                                                                                          |              |                |             |
| 2            | What are the main limitations or challenges when using Boundary Value Analysis? And when would you typically apply BVA during the testing process?                                        | 8            | Section-3      | Unit-3      |
| 3            | What is Equivalence Partitioning (EP) in software testing, and what is its primary goal? And how does Equivalence Partitioning complement Boundary Value Analysis (BVA)?                  | 8            | Section-3      | Unit-3      |
| 4            | What is Decision Table Testing, and when is it particularly useful? And what are the main components of a Decision Table? Explain each briefly.                                           | 8            | Section-3      | Unit-3      |
| 5            | What is State Transition Testing, and when is it most applicable and its advantages?                                                                                                      | 8            | Section-3      | Unit-3      |
| 6            | What are the key components of a State Transition Diagram, which is often used in State Transition Testing?                                                                               | 8            | Section-3      | Unit-3      |
| 7            | How do you design test cases from Use Cases?                                                                                                                                              | 8            | Section-3      | Unit-3      |
| 8            | List and briefly describe the common states in a typical Defect Life Cycle.                                                                                                               | 8            | Section-3      | Unit-3      |
| 9            | List at least five key features or functionalities typically found in a good Bug Tracking System.                                                                                         | 8            | Section-3      | Unit-3      |
| 10           | What is the difference between "Severity" and "Priority" in bug tracking?                                                                                                                 | 8            | Section-3      | Unit-3      |
| <b>Q.no.</b> | <b>Questions (Unit-4)</b>                                                                                                                                                                 | <b>Marks</b> | <b>Section</b> | <b>Unit</b> |
| 1            | a) What is Test Automation, and what is its primary goal in software testing?<br>b) Differentiate between Manual Testing and Automation Testing. Provide at least three key differences.  | 8            | Section-4      | Unit-4      |
| 2            | a) List at least five benefits of implementing Test Automation in a software development project.<br>b) What types of testing are generally well-suited for automation? Provide examples. | 8            | Section-4      | Unit-4      |
| 3            | What is a Test Automation Tool, and give examples of popular tools for web application testing? What are some common challenges or disadvantages associated with Test Automation?         | 8            | Section-4      | Unit-4      |
| 4            | a) What is a Test Automation Framework, and why is it important to use one?<br>b) Describe three common types of Test Automation Frameworks and their distinguishing characteristics.     | 8            | Section-4      | Unit-4      |
| 5            | Why is it important to consider the "domain" or type of application when recommending an automation framework or tool?                                                                    | 8            | Section-4      | Unit-4      |
| 6            | State and explain advantages and disadvantages of Domain Testing.                                                                                                                         | 8            | Section-4      | Unit-4      |
| 7            | You need to set up automation for a desktop application (e.g., a Windows-based application). What challenges might you face, and what tool categories would you consider?                 | 8            | Section-4      | Unit-4      |
| 8            | For mobile application testing (IOS and Android native apps), what factors influence tool and framework selection, and what common approaches are used?                                   | 8            | Section-4      | Unit-4      |
| 9            | What is the Process of Automation testing? Explain with examples.                                                                                                                         | 8            | Section-4      | Unit-4      |
| 10           | What is the purpose of Reporting in Test Automation, and what key information should a good automation report provide?                                                                    | 8            | Section-4      | Unit-4      |

| Q.n<br>o. | Questions (Unit-5)                                                                                                                   | Marks | Section   | Unit   |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|--------|
| 1         | What are the key components typically required for setting up an automation testing environment for web applications using Selenium? | 8     | Section-5 | Unit-5 |
| 2         | Why is it recommended to use a Build Automation Tool like Maven or Gradle in a Selenium project?                                     | 8     | Section-5 | Unit-5 |
| 3         | What are the common issues faced during environment setup for Selenium automation, and how can they be mitigated?                    | 8     | Section-5 | Unit-5 |
| 4         | What is Selenium WebDriver, and how does it interact with web browsers to perform automation?                                        | 8     | Section-5 | Unit-5 |
| 5         | What is TestNG, and what are its main advantages when used with Selenium WebDriver?                                                  | 8     | Section-5 | Unit-5 |
| 6         | Explain the purpose of @BeforeMethod and @AfterMethod annotations in TestNG, and how they are typically used in Selenium automation. | 8     | Section-5 | Unit-5 |
| 7         | How can you perform data-driven testing using TestNG in a Selenium project? Illustrate with a simple example structure.              | 8     | Section-5 | Unit-5 |
| 8         | Describe the process of setting up and running a TestNG test suite using testng.xml                                                  | 8     | Section-5 | Unit-5 |
| 9         | How does Maven contribute to the implementation and management of a Selenium automation framework?                                   | 8     | Section-5 | Unit-5 |
| 10        | What is Git, and why is it essential for collaborative development and framework implementation in an automation project?            | 8     | Section-5 | Unit-5 |